

PRODUCT DATA SHEET

VIR TOPAZ (CGS / CGF)



DESCRIPTION

VIR TOPAZ Compact Laminate is designed for premium interior surfaces such as kitchen tops, worktops, and countertops. Made by compressing multiple layers of phenolic Kraft paper and décor paper with melamine resin, it features a melamine-impregnated overlay that enhances durability and surface protection. Manufactured in compliance with EN 438-4 and EN 13501-1, TOPAZ delivers reliable quality, safety, and long-lasting performance for everyday use.

Description: VIR TOPAZ

Laminate Grade 1: CGS (Compact General Purpose Standard)

Laminate Grade 2: CGF (Compact General Purpose Fire retardant)

Sizes: 1830x1830 mm, 1830x2165 mm, 1830x3660 mm, 1830x4330 mm

Thickness: 10 mm – 18 mm

APPLICATION / APPLICATION AREA

As Kitchen Tops, Work Tops, Counter Tops for general usage.



MIGRATION TESTING

VIR Topaz for kitchen worktops are tested in accordance with EN 1186 and EN 13130, with reference to Regulation (EU) No 10/2011, to evaluate the migration of substances from the laminate surface when in contact with food.

1. OVERALL MIGRATION LIMIT

Tested as per EN 1186 to measure the total amount of substances that may migrate from the compact laminate surface into food simulants.

Simulants used	Test condition	Result (mg/dm ²)			Reporting Limit (mg/dm ²)	Permissible Limit (mg/dm ²)
		1st Migration	2nd Migration	3rd Migration		
3% Acetic Acid	70°C For 02 Hours	9.06	3.07	Not Detected	2	10
95 % Ethanol	60°C For 02 Hours	Not Detected	Not Detected	Not Detected	2	10
Isooctane	60°C For 02 Hours	Not Detected	Not Detected	Not Detected	2	10
Stability^		Yes	Yes	Yes	-	-
Conclusion		Pass	Pass	Pass	-	-

2. SPECIFIC MIGRATION LIMIT

Tested in accordance with EN 13130 to evaluate the migration of specific individual substances from the laminate into food simulants.

Test Name	Standard	Simulants	Unit	Permissible Limit	Reporting Limit	Conclusion
Specific Migration of Melamine	EN 13130-1	3% Acetic acid	mg/kg	2.5	0.5	Not Detected
Specific Migration of Formaldehyde	EN 13130-1	3% Acetic acid	mg/kg	15	3	Not Detected
Specific Migration of Phenol	EN 13130-1	3% Acetic acid	mg/kg	3.0	0.5	Not Detected
Specific Migration of Bis-Phenol A	EN 13130-1	3% Acetic acid	mg/kg	0.001	0.001	Not Detected
SSpecific Migration of PAA's	EN 13130-1	3% Acetic acid	mg/kg	0.01	0.002	Not Detected
Specific Migration of Phthalates	EN 13130-1	95% Ethanol	mg/kg	0.6	0.1	Not Detected

SENSORY EXAMINATION – ODOUR AND TASTE TEST

To comply with Odour and taste requirements, VIR TOPAZ was subjected to sensory examination in accordance with DIN 10955: 2024-1.

Test Media	Test Item	Result	Maximum Permissible Limit
Edible oil	Sensorial examination Odour	0	2.5
	Sensorial examination Taste	0	2.5
Conclusion		Pass	-

The sample exhibited an intensity score of 0 for both Odour and Taste, against the permissible limit of 2.5, indicating no perceptible sensory deviation.

U.S. FDA FOOD CONTACT COMPLIANCE

VIR TOPAZ for kitchen worktops are evaluated in accordance with applicable **U.S. Food and Drug Administration (FDA)** regulations for food contact materials to assess suitability for repeated food contact applications.

Test Name	Standard	Test / Extractant Media	Test Condition	Unit	Permissible Limit	Detection Limit	Result	Conclusion
Determination of Amount of Extractives - Polymeric or Resinous Coatings	US FDA 21 CFR 175.300	Distilled Water	120°F for 24 Hours	mg/in ²	18	1	Not Detected	Pass
		8% Alcohol	120°F for 24 Hours	mg/in ²	18	1	Not Detected	Pass
		n-Heptane	70°F for 30 Minutes	mg/in ²	18	1	Not Detected	Pass
Chloroform Soluble Extractives – Melamine formaldehyde surfaces	US FDA 21 CFR 177.1460	Distilled Water	120°F for 24 Hours	mg/in ²	0.5	1	Not Detected	Pass
		8% Alcohol	120°F for 24 Hours	mg/in ²	0.5	1	Not Detected	Pass
		n-Heptane	70°F for 30 Minutes	mg/in ²	0.5	1	Not Detected	Pass

REACH 253 SVHCS CANDIDATE LIST SCREENING TEST

VIR TOPAZ is tested in accordance with the requirements of **REACH Regulation (EC) No. 1907/2006** to evaluate the presence of Substances of Very High Concern (SVHCs) included in the European Chemicals Agency (ECHA) Candidate List of 253 substances.

Chemical Category	Key Regulated Substances Covered	Regulatory Basis	Observed Value
Phthalates / Plasticizers	DEHP, DBP, BBP, DIBP, DCHP	REACH SVHC – Reprotoxic Substances	PASS – ≤ 0.1% (w/w)
Heavy Metals & Metal Compounds	Lead, Cadmium, Lead compounds, Cadmium compounds, Chromium VI compounds	REACH SVHC – Toxic / Carcinogenic Metals	PASS – Not Detected / ≤ 0.1% (w/w)
Bisphenols & Phenolic Compounds	Bisphenol A (BPA), Nonylphenol, Octylphenol derivatives	REACH SVHC – Endocrine Disruptors	PASS – Not Detected / ≤ 0.1% (w/w)
Per- & Polyfluoroalkyl Substances (PFAS)	PFOA, PFNA, PFDA, PFHxS related substances	REACH SVHC – Persistent Bio accumulative Chemicals	PASS – Not Detected / ≤ 0.1% (w/w)
Polycyclic Aromatic Hydrocarbons (PAHs)	Benzo[a]pyrene, Chrysene, Fluoranthene, Phenanthrene, Pyrene	REACH SVHC – Carcinogenic Substances	PASS – Not Detected / ≤ 0.1% (w/w)
Organotin Compounds	DOTe, Organotin derivatives	REACH SVHC – Toxic Organometallic Compounds	PASS – Not Detected / ≤ 0.1% (w/w)
Azo Dyes & Aromatic Amines	Carcinogenic azo substances and aromatic amines	REACH SVHC – Carcinogenic / Mutagenic Substances	PASS – Not Detected / ≤ 0.1% (w/w)
Flame Retardants & Chlorinated Compounds	HBCDD, SCCP, Brominated / Chlorinated substances	REACH SVHC – PBT / vPvB Chemicals	PASS – Not Detected / ≤ 0.1% (w/w)
Other Substances of Very High Concern (SVHC)	Borates, solvents, sensitizers and other listed SVHC substances	Regulation (EC) No. 1907/2006 (REACH) – 253 Candidate List	PASS – All tested substances ≤ 0.1% (w/w)

Note: The above listed substance groups represent the principal categories covered under the REACH SVHC Candidate List (253 substances) in accordance with Regulation (EC) No. 1907/2006 (REACH). The screening scope covers all applicable 253 Substances of Very High Concern (SVHC) as published by the European Chemicals Agency (ECHA). A complete detailed chemical test report covering all individual SVHC substances is available upon request.

FIRE PERFORMANCE (ON DEMAND)

VIR TOPAZ (CGF grade) is CE certified for interior applications, demonstrating compliance with the Construction Products Regulation (CPR). The product meets fire performance classification requirements as per EN 13501-1. Fire testing has been carried out in accordance with EN 13823:2020 + A1:2022 to validate its reaction to fire performance.

Reaction to Fire Classification: Classified as Class B, demonstrating excellent fire resistance with a Fire Growth Rate FIGRA0.2MJ (W/s) and FIGRA0.4MJ (W/s) of only 26.1 W/s and 18.3 W/s and a Total Heat Release (THR_{600s}) of 2.4 MJ.

Smoke Development: Rated s1, indicating minimal smoke emission. The Smoke Growth Rate (SMOGRA) is 1.5, and the Total Smoke Production (TSP_{600s}) is just 27.7 m².

Flaming Droplets / Particles: Rated d0, confirming no flaming droplets or particles ≥10 seconds within the first 600 seconds of testing.

Parameter	Requirements of B-s1,d0 Class of EN 13501- 1:2018	Observed Results	Conformity (Confirms / Do not Confirms)
A. Single Burning Item Test			
FIGRA0.2MJ (W/s)	≤120 W/s	26.1	Confirms
FIGRA0.4MJ (W/s)	≤120 W/s	18.3	Confirms
THR _{600s} (MJ)	≤7.5 MJ	2.4	Confirms
SMOGRA (m ² /s ²)	≤30 m ² /s ²	1.5	Confirms
TSP _{600s}	≤50 m ²	27.7	Confirms
LFS to edge	No LFS to edge	No	Confirms
Flaming Droplet / Particles ≥10s within the first 600 seconds	No	No	Confirms
B. Ignitability Test			
Flame to reach 150mm Mark	No Flame should reach 150mm mark	No Flame reached 150mm mark	Confirms

REACTION TO FIRE CLASSIFICATION: B-s1, d0



EN 13823:2010
EN ISO 11925-2:2010

QUALITY CHARACTERISTICS / TECHNICAL DATA

TOPAZ of both General purpose and Flame- retardant (CGS/CGF) are classified under EN-438-4 and tested in accordance to EN-438-2 to verify their quality, durability, and safety.

PROPERTIES	TEST METHOD	ATTRIBUTES OF PERFORMANCE	UNIT OF MEASUREMENT	MINIMUM REQUIREMENT	RESULTS
Surface Quality	EN 438-2.4	Stain, Dirt and Similar defects	mm ² /m ²	≤ 2	1
		Fibers, hairs and stripes	mm ² /m ²	≤ 20	10
Resistance to Surface wear	EN 438-2.10	Wear Resistance - Initial point	Revolutions until IP (min)	≥ 150	450
Resistance to immersion in boiling water	EN 438-2.12	Mass increases - T≥5 mm	% (max)	2	0.4
		Thickness increases - T≥5 mm	% (max)	2	0.2
		Appearance - Gloss finish	Rating	≥ Grade 3	5
		Appearance - Other finish	Rating	≥ Grade 4	5
Resistance to water vapour	EN 438-2.14	Appearance - Gloss finish	Rating	≥ Grade 3	5
		Appearance - Other finish	Rating	≥ Grade 4	5
Resistance to Moisture	EN 438-2.15	Mass increases - T≥5 mm	% (max)	2	0.2
		Appearance	Rating	≥ Grade 4	5
Resistance to dry heat (160°)	EN 438-2.16	Appearance - Gloss finish	Rating	≥ Grade 3	5
		Appearance - Other finish	Rating	≥ Grade 4	5
Dimensional stability at elevated temperatures	EN 438-2.17	Cumulative dimensional change - T ≥ 5 mm	L % (max)	0.3	0.1
			T % (max)	0.6	0.08
Resistance to impact with large diameter ball	EN 438-2.21	T ≥ 6 mm Drop height >1800mm	Mm	Indent. Dia. 10mm	5
Resistance to Crazing	EN 438-2.24	Appearance	Rating	≥ Grade 4	5
Resistance to Scratching	EN 438-2.25	Appearance - Smooth Finishes	Rating	≥ Grade 2	5
		Appearance - Textured Finishes	Rating	≥ Grade 3	5
Resistance to Staining	EN 438-2.26	Appearance - Group 1 &2	Rating	= Grade 5	5
		Appearance - Group 3	Rating	≥ Grade 4	5
Light Fastness (Xenon arc)	EN 438-2.27	Contrast	Grey scale rating	4-5	5
	ASTM G53-91 (314-400nm)	Contrast	Wool Scale rating	≥ 4	6
Modulus of elasticity	EN ISO 178	Stress	Mpa (min)	≥ 9000	15336
Flexural strength	EN ISO 178	Stress	Mpa (min)	≥ 80	136.1
Density	EN ISO 1183	Density	gm/cm ³	≥ 1.35	1.47
Resistance to Fixing	ISO 13894-1	Screw Pullout force	N	-	3514
Thermal Conductivity	EN 12524	Thermal Conductivity	W/mK	≤ 0.30	0.2
Antibacterial Activity	JIS Z 2801: 2010	Log Reduction	%	R > 2	>99.99
Antiviral Activity	ISO 21702: 2019	Reduction rate	%	-	>99.99
Antifungal Activity	ASTM G 21 - 21	Reduction rate	Rating	≤1	0
Formaldehyde emission	EN 16516	Emission Level	mg/m ³	≤ 0.124	0.01
	EN 717-2	Classification	Class	-	E1

SURFACE FINISH

The surface finish test is conducted in accordance with EN 13722 to evaluate the angle of light reflection for each surface finish. The results are expressed in gloss units (GU), providing a precise measure of the reflective characteristics.

SURFACE	MAXIMUM DEVIATION	GLOSS UNITS
High Gloss Surface	± 15	> 70
Semi-Gloss Surface	± 10	30 - 70
Suede Surface	± 5	10 - 30
Matt surface	± 3	< 10

DIMENSIONAL TOLERANCES

NOMINAL THICKNESS (mm)	THICKNESS TOLERANCE (mm)	LENGTH & WIDTH (mm)	STRAIGHTNESS OF THE EDGES (mm/m)	SQUARENESS (mm/m)	FLATNESS (mm/m)
2-3	±0.2	+10/-0	≤ 2	≤ 1.5	≤ 2
3-5	±0.3	+10/-0	≤ 2	≤ 1.5	≤ 2
5-8	±0.4	+10/-0	≤ 2	≤ 1.5	≤ 2
8-12	±0.5	+10/-0	≤ 2	≤ 1.5	≤ 2
12-16	±0.6	+10/-0	≤ 2	≤ 1.5	≤ 2
16-20	±0.7	+10/-0	≤ 2	≤ 1.5	≤ 2
20-25	±0.8	+10/-0	≤ 2	≤ 1.5	≤ 2